

Ivana Seric

New Jersey Institute of Technology
Newark, New Jersey
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RESEARCH INTERESTS

Computational fluid dynamics, mathematical modeling, ferrofluids, basketball analytics

EDUCATION

Doctor of Philosophy, Mathematical Sciences
New Jersey Institute of Technology, Newark, NJ, expected May 2016
Concentration: Applied Mathematics

Online specialization: Data Science
Johns Hopkins University, Baltimore, MD

Bachelor of Science, Mathematical Sciences
New Jersey Institute of Technology, Newark, NJ, May 2011
Concentration: Applied Mathematics

PUBLICATIONS

I. Seric, S. Afkhami, and L. Kondic, "Interfacial instability of thin ferrofluid films under a magnetic field", *Journal of Fluid Mechanics Rapids*, vol. 755(9), 2014.

S. P. Naughton, N. K. Patel, I. Seric, "Instability of gravity driven flow of liquid crystal films". *SIAM Undergraduate Research Online (SIURO)*, vol. 5, 2012.

CONFERENCE TALKS

I. Seric, K. Mahady, S. Afkhami, and L. Kondic, "Volume of fluid simulations of liquefied metal nanofilms with Marangoni effects", American Physical Society, 67th Annual Meeting of the Division of Fluid Dynamics, San Francisco, CA, November 23-25, 2014.

I. Seric, S. Afkhami, and L. Kondic, "Numerical simulations of a ferrofluid drop on a substrate under an applied magnetic field", American Physical Society, 66th Annual Meeting of the Division of Fluid Dynamics, Pittsburgh, PA, November 24-26, 2013.

POSTERS

I. Seric, S. Afkhami, and L. Kondic, "Interfacial instability in thin ferrofluid films under a magnetic field", *Frontiers in Applied and Computational Mathematics 2014*, New Jersey Institute of Technology, Newark, NJ, May 22-23, 2014.

I. Seric, S. Afkhami, and L. Kondic, "Interfacial instability in thin ferrofluid films under a magnetic field", *Dana Knox Student Research Showcase 2014*, New Jersey Institute of Technology, Newark, NJ, April 16, 2014.

I. Seric, S. Afkhami, and L. Kondic, "Numerical simulations of a ferrofluid drop on a substrate under an applied magnetic field", *Graduate Student Research Day 2013*, New Jersey Institute of Technology, Newark, NJ, October 31, 2013.

I. Seric, S. Afkhami, and L. Kondic, "Long-wave approximation of a ferrofluid film under an external magnetic field", *Frontiers in Applied and Computational Mathematics 2013*, New Jersey Institute of Technology, Newark, NJ, May 31 - June 2, 2013.

TEACHING EXPERIENCE

Substitute instructor
New Jersey Institute of Technology, Newark, NJ

Fall 2014 - Present

	<i>Lab instructor</i> Math 340 - Applied Numerical Methods New Jersey Institute of Technology, Newark, NJ	Spring 2014
	<i>Instructor</i> Math 132 - Calculus B New Jersey Institute of Technology, Newark, NJ	Fall 2013
	<i>Recitation instructor</i> Math 111, Math 112 Honors Math 111, Math 111 Honors Math 112, Math 112 Honors New Jersey Institute of Technology, Newark, NJ	Spring 2012 Fall 2012 Spring 2013
	<i>MATLAB Helpdesk</i> New Jersey Institute of Technology, Newark, NJ	Fall 2011
	<i>Math and Physics Tutor</i> Center for Academic and Personal Enrichment New Jersey Institute of Technology, Newark, NJ	June 2008 - May 2010
TECHNICAL SKILLS	<i>Programming Languages & Software:</i> C, C++, R, Python, MPI, OpenMP, FORTRAN, MATLAB, Mathematica, Bash, HTML. <i>Operating Systems:</i> Linux, Windows.	
LANGUAGES	English (bilingual proficiency) Croatian (native) German, Russian (basic proficiency)	
RESEARCH EXPERIENCE	<i>Research assistant</i> Computational Fluid Dynamics with applications to magnetohydrodynamics and liquid metal thin films. Developed a mathematical model and implemented a computational numerical model.	Spring 2012 - Present
RESEARCH WORKSHOPS	<i>Modeling Problems in Industry 2014</i> , Worked on "Effects of membrane morphology on separation efficiency"; problem presented by Dr. Mark Hurwitz, Pall Corporation. New Jersey Institute of Technology, Newark, NJ, June 23 - 27, 2012. <i>Modeling Problems in Industry 2012</i> , Worked on "Modeling industrial coalescers: droplet dynamics"; problem presented by Dr. Mark Hurwitz, Pall Corporation. University of Delaware, Newark, DE, June 11 - 15, 2012. <i>Graduate Student Math Modeling Camp 2012</i> , Worked on "Modeling Sterling Engine"; problem presented by Dr Sean Bohun. Rensselaer Polytechnic Institute, Troy, NY, June 5 - 8, 2012.	
PROFESSIONAL ORGANIZATIONS	Pi Mu Epsilon - Mathematical Honor Society. American Physical Society Society for Industrial and Applied Mathematics	

**BASKETBALL
EXPERIENCE
AND
AWARDS**

Member of NJIT women's basketball team, NCAA Division I *Spring 2008 - 2011*
Team leader in rebounds, assists, steals, and blocks in the senior year.
Started every game but one during college career.
Team Leading Rebounder Award 2010/2011.
Team Best Assist Leader Award 2008/2009.
Team Most Inspired Player Award 2007/2008.
NJIT Division I career total record: assists, blocks, FG percentage.
NJIT Division I single-season record: assists (2010/2011), blocks (2010/2011), FG percentage (2007/2008).
NJIT Division 1 single-game record: rebounds (2011), blocks (2011), assists (2010).
Great West conference honorable mention 2011.
Great West conference All-Academic Team 2009, 2010, 2011.

Member of Croatian national U16 (2003, 2004) and U18 (2005, 2006) teams.
Bronze medal in European championship U18 - division B in 2006.

Croatian national youth competitions (club competitions):
U14: Bronze medal (BC Split, 2001).
U16: Silver medal U16 (BC Split, 2003).
U18: Bronze medal (BC Split, 2004), two silver medals U18 (BC Pozega 2005, 2006).

School competitions
High School national championship: Bronze medal (Gimnazija Pozega 2007)
High School city championship: Silver medal (III Gimnazija Split 2004)
Primary School city championship: Gold medal (OS Blatine-Skrape, 2000, 2001, 2003).
Primary School regional championship: Silver medal (OS Blatine-Skrape, 2000), Bronze medal (OS Blatine-Skrape, 2001).

HOBBIES

Running, Cycling, Basketball, Hiking